Using State of the Art Feature Annotation to Predict the Pathogenicity of coding variants in Cancer

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1 Introduction

Genome sequencing determines the complete DNA sequence of an organism. To identify variations in a genome, a sequenced sample is aligned to a reference genome, a genome which acts as a baseline for comparison. Variant calling is the process of identifying the differences between the sample genome and the reference genome. Specialised algorithms scan the alignment to detect variation, examples of variants include:

- Single Nucleotide Variants (SNV): Single base changes, e.g. A to G
- Insertion and Deletions (INDELs): Small insertions or deletions of bases

Variants are annotated to add biological context to each variant, linking it to genes, proteins and functions. Databases like COSMIC [17], for cancerous mutations, help classify variants as pathogenic, benign or uncertain significance based on known associations with disease. In bioinformatics, the vast amount of genomic sequencing data can be computationally analysed to extract meaningful insights into genetic function, disease mechanisms and evolutionary relationships. A key challenge is the prediction of a variant's driver status in relation to cancer. A variant can either "drive" cancer development, or be a "passenger" representing random mutations which don't drive cancer. Identifying drivers could provide new insights into the molecular mechanisms underpinning tumor progression. This introduces novel biomarkers or therapeutic targets for cancer treatment [7].

To determine the driver status of indels within the coding region of the genome, a subset of indel mutations are considered. In coding regions, DNA is read in sets of three bases (codons), each coding for a specific amino acid in a protein. Frameshift indels are insertions or deletions to the genomic sequence which are not multiples of 3 bases. They shift the sequence in which the codons are read. This is known as shifting the reading frame. Subsequently, codons are read incorrectly which can introduce unintended "stop" codons earlier than normal. This results in a truncated protein, which may not perform it's intended function [2]. In-frame indels are insertions or deletions which are multiples of 3 bases, they do not disrupt the reading frame. They might be less disruptive than frameshift indels, but some are still thought to play a big role in driving disease. An example of a cancer-driving coding indel occurs in the APC gene, where it causes a frameshift mutation that introduces a premature stop codon, resulting in loss of gene function. This loss of APC function leads to uncontrolled cell proliferation, contributing to the development of colorectal cancer [9].

Classifying the driver status of cancer-related coding indels is under-researched in comparison with SNVs. Indels have more complex structural and functional effects due to their protein length altering nature, making it harder to determine their impacts based on features used by SNV classifiers. This project aims to build and benchmark a model which accurately classifies the driver status of such indels. To achieve higher performance metrics than pre-existing models, it will leverage the state of the art tool, Drivr-Base, which generates a comprehensive and extensive compilation of feature annotations across the entire genome [4].

2 Literature review

2.1 Existing models

Several computational models in the bio-informatics field have been developed to classify the pathogenicity of indels, this is accomplished by distinguishing between driver mutations and neutral mutations. FATHMM-Indel and SIFT-Indel focus on different mutation contexts and goals. FATHMM-Indel assesses the functional impact of small indels within the non-coding region of the genome, leveraging conservation-based features obtained from multiple sequence alignments (MSAs) of vertebrate genomes to human genomes [14]. The model achieves an 89% testing accuracy on an independent test set, underscoring its ability to accurately classify non-coding indels [15].

In contrast, SIFT-Indel, an extension of the SIFT Algorithm [16] which predicts the effects of protein function, addresses frameshift indels which occur within coding regions. SIFT-Indel uses a J48 decision tree algorithm because of the clear interpretability of classification rules it provides. It uses a training dataset composed of disease-causing frameshift indels from the Human Gene Mutation Database (HGMD) and neutral indels from the pairwise alignments from the UCSC genome browser [1]. SIFT-Indel achieves an overall accuracy of 84% and a sensitivity of 90% [8].

Another model, PredCID also distinguishes between driver and passenger frameshift indels, but differing with the scope of disease revolving around cancer. PredCID employs an XGBoosting algorithm to classify the indels, exhibiting extremely high performance metrics, outperforming existing tools in driver indel prediction [18]. A possible reason for this is their inclusion of PubMed features, which can impose bias into the model.

2.2 Feature Annotation

Feature annotation is a core component in machine learning models, features provide the necessary biological information attributed to the indel's genomic position relating to the driver status. It is paramount each indel has sufficient information so models can generalise well to new data.

David Woodruff (2004) proposed that in areas of high conservation, mutations have greater consequences [19]. Centering the feature annotation around conservation should provide insights into the likelihood of an indels deleterious impact based on how conserved the surrounding sequence is. FATHMM-Indel's feature selection focuses on conservation-based metrics, particularly PhyloP and PhastCons scores, which assess evolutionary constraint around the mutation site. Conservation scores are accessed through the University Of California Santa Cruz (UCSC) Genome Browser [1,3,11]. FATHMM-Indel's high accuracy is a testament to the importance of conservation features for classification.

For models focusing on coding regions, SIFT-Indel and PredCID, feature sets are expanded to include structural and functional aspects relevant to frameshift indels. PredCID introduces features spanning gene, DNA, transcript and protein levels. Gene-level features assess a gene's cancer relevance via the frequency of PubMed mentions created by VEST-Indel; DNA-level features capture indel length and conservation; transcript-level features consider relative indel location and downstream codon impact; and protein-level provides a measure of how much a protein's domain structure might be impacted by the indel mutation, quantifying the proportion of domains affected by the indel [18]. PubMed features reflect on prior research focusing on specific genes. The frequency of PubMed mentions could correlate with a pathogenic outcome, as disease-related genes tend to be more heavily studied. This introduces a bias, as the model may use the frequency of research attention as a proxy for pathogenicity rather by learning from the biological attributes alone. PredCID's use of PubMed features in their model warrants a careful analysis into this bias before implementation. SIFT-indel has 20 features, but narrows them down to 4 dominant features which achieve a better classification performance. These consist of 2 DNA level features similar to PredCID's, and 2 additional transcript level features; minimum distance of an indel to the exon boundary of all affected transcirpts, and indel location relative to the transcript [8].

2.3 Modelling Techniques

Various machine learning models have been used in these predictive tools, each chosen based on the complexity of the data and interpretability requirements. FATHMM-Indel uses Support Vector Machines (SVMs) [5] for a binary classifier because of their robust ability to handle structured biological data and popular use across the bioinformatics field. In contrast, SIFT-Indel uses a decision tree classifier, to utilise its transparent decision-making process to provide interpretable results about the functional impact of frameshift indels.

PredCID setup multiple machine learning models (XGBoost, Random Forest, Decision Tree, Random Tree, Adaboost and Naive Bayes) and evaluated their performances. It was deduced that XGBoost, a relatively new method, had the best performance metrics where precision, sensitivity, specificity and MCC are 0.969, 0.974, 0.969 and 0.943, respectively. PredCID achieved a higher AUPR and AUC on the independent test set, demonstrating superior predictive power over competing tools such as FATHMM-Indel, CADD and SIFT-Indel, especially in assessing cancer driver indels. However, considering PredCID's use of Pubmed features at the gene level, these performances are likely inflated due to research density bias.

2.4 Feature Engineering

A state of the art feature annotation tool for SNVs, DrivR-Base, "generates a comprehensive and extensive compilation of annotations across the entire genome. It has unparalleled breadth and depth of genomic and protein-level information" [4]. DrivR-Base offers access to a diverse range of annotations across the entire genome, sourced from multiple public databases. These annotations include conservation metrics like PhyloP and PhastCons, regulatory features from ENCODE, and structural properties from AlphaFold, among others [3, 6, 10, 11]. These features combined could allow high dimensional machine learning models to classify indels more accurately. Therefore, it is possible that by adapting Drivr-Base for indel annotations could boost classification performances to greater than 90% , surpassing current standards.

2.5 CanDrivr-INDEL

The models discussed above all have one limitation in common, a constrained small feature set. Combining the strong features with an adapted version of DrivR-Base for indels will provide an extensive and valuable feature set for indels. Leveraging the new XGboost classifier, pro

3 Plan

Table 1: Project Plan

Weeks	**Action**
4	Read Papers: FATHMM-Indel, CADD, Drivr-Base. Understand what sort of data they use for their models, and locate where I can get my data from
5	Use Drivr-Base, learn how to use docker, understand how the features are annotated to each variant. Understand what the features are and their importance. Get an example working.
6	Adapt Drivr-Base for Indels and get conservation and dna shape features.
7	Research databases to get a subset of neutral “Passenger” Indels for the classification. Set up the first model using xgboost classifier and get some results for the interim report
8	Analyse results, remove any bias which is occuring because of annotation
9-10	Send first draft of interim report to supervisor for some feedback. Then submit final draft in week 10.
11-12	Collect and curate a better, larger driver and passenger dataset. Finalise the dataset I will be using for modelling. Also create an independent test set of indels for model performance evalutation - ensure 0 circularity.
Christmas Break	
13	Develop Drivr-Base-INDEL, adapting DrivR-Base to annotate all the features for Indels instead of SNVs.
14-15	Assess a range of classifiers. Add the pubmed feature which has been associated with over-performing in previous papers. Run models again including this feature, comparatively analyse the results. Conclude if this feature has any impact on performance
16-17	Develop a confidence measure for the best performing model (Likely xgboost). Start writing draft of the 1st chapter
18-19	Investigate the counts of Indels inside the cancer genome for specific cancers in relation to Darbyshire et al paper.
20	Make the code for the model and DrivrBase-Indel reproducable and usable for other researchers. Share draft of 1st chapter.
21-22	Begin making the poster, write a draft of the 2nd chapter.
23-24	

4 Progress

To prioritise the modeling process, the datasets are kept small due to the time-intensive process of assigning features.

4.1 Data Collection

ClinVar, a publicly accessible database, archives human genetic variants and their associations with phenotypes, classified as pathogenic (disease-causing), benign (not disease-causing), or of uncertain significance [13]. ClinVar was filtered for frameshift and in-frame insertion/deletion variants of base length smaller than 20, labeled as benign. Most cancers are driven by mutations on autosomes (chromosomes 1 through 22), as this is where most of the proto-oncogenes (cell growth regulating genes) and tumor suppressor genes reside, therefore sex chromosomes X and Y are omitted. This filtering yielded a dataset of 2,458 indels spanning all autosomes, with these variants labeled as "passenger" (label = 0). To collect cancer-related indels COSMIC, the largest repository of somatic cancer mutations, was used [17]. COSMIC's Gene Mutation Census data were filtered to select frameshift and in-frame indels across autosomes, resulting in a comparably sized dataset. These variants were labeled as "driver" mutations (label = 1).

4.2 Feature Annotation

Adapting Drivr-Base to retrieve conservation scores for each indel's genomic position span resulted in a set of conservation measures: 4, 7, 17, 20, 30, 100, 470 way phyloP and phastCons scores [3, 11] , and k24, k36, k100 Umap and Bismap scores for quantifying genome and methylome mappability [12]. For insertion variants, scores were obtained by averaging the scores of adjacent positions, whereas deletion scores were computed by averaging scores across the deleted positions. DNA shape metrics, adapted for indels, were generated using DrivR-Base [4]. These metrics include characteristics: minor groove width, Roll, propeller twist, helix twist and electrostatic potential. These gain insights into how the indels impact the structural properties of DNA. In total these annotations initially produced 102 metrics. However, some DNA shape metrics included NaN values exclusively within the benign dataset, introducing bias. To avoid classifier bias based on missing values, columns containing these NaN values were removed, resulting in a final set of 90 annotations.

4.3 Classification

The training and testing sets are created using an 80:20 randomised split. A binary logistic classification XGboost classifier is set up with logloss evaluation metrics. The logistic approach is used so the output is a probability that a given input belongs to the positive class. Leave One Chromosome Out (LOCO) Cross Validation ensures that the training and testing sets are separated, reducing the risk of overfitting by forcing the model to learn generalisable patterns across all chromosomes rather than memorising specific details from the training data.

5 Results

The XGBoost classifier is evaluated on the test set to assess its predictive performance for classifying driver and passenger indels. The following classification report in Table 2 and visualisations were generated to assess its effectiveness. The classification report summarises the performance of a binary classifier, where the two classes, passengers and drivers, are labeled as 0 and 1.

Table 2: Classification Report and Accuracy Metrics

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0	0.78	0.76	0.77	494
1	0.79	0.81	0.80	552
Accuracy	0.78 (1046)			
Macro avg	0.78	0.78	0.78	1046
Weighted avg	0.78	0.78	0.78	1046

Metric	**Value**
Mean LOCO-CV Accuracy	0.7988
Test Set Accuracy	0.7820

Precision, Recall, and F1-Score

- **Precision**: Measures the proportion of true positive predictions out of all predictions made for a particular class. It answers the question, *"Of all the indels predicted as driver, how many are actually correct?"*

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

- **Recall**: Recall measures the proportion of true positive predictions out of all actual instances of a particular class. It answers the question, *"Of all the actual instances of a class, how many did the model correctly identify"*

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

- **F1-Score**: The F1-score is the harmonic mean of precision and recall. It provides a single metric that balances both.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Support represents the number of true instances for each class in the test data. For passengers, support is 494, and for drivers, support is 552.

Accuracy

Accuracy is the overall proportion of correct predictions made by the classifier.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

Macro and Weighted Averages

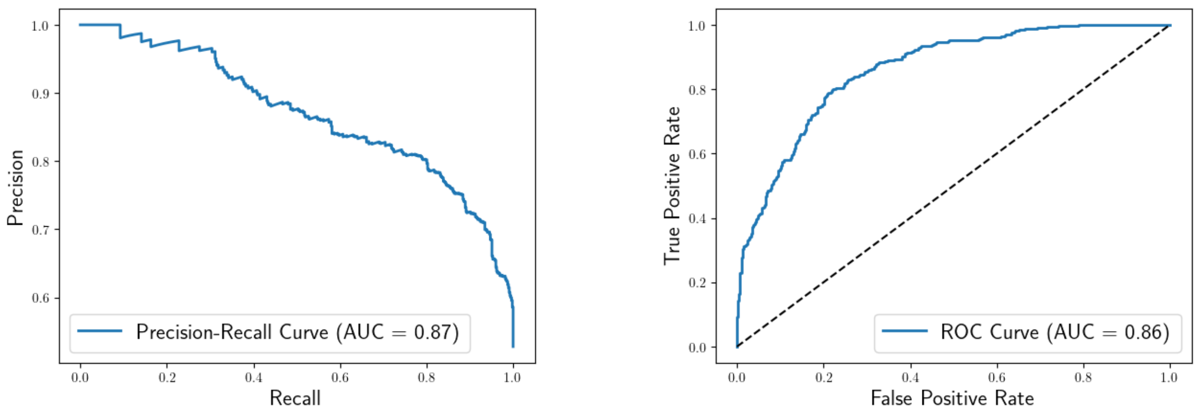
- **Macro Average**: This is the average of precision, recall, and F1-score calculated independently for each class, treating all classes equally (without considering class frequency).

$$\text{Macro Precision} = \frac{\text{Precision}_{\text{passenger}} + \text{Precision}_{\text{driver}}}{2} \tag{5}$$

- **Weighted Average**: This is the average of precision, recall, and F1-score for each class, weighted by the support (number of instances) of each class. This average considers class frequency, making it more representative of real-world performance when classes are imbalanced.

The performance metrics in Table 2 present good results with an accuracy of 78% on the test set and similar metrics across precision, recall, and F1-score. The close alignment of test and LOCO cross-validation accuracies indicates reasonable generalisation to unseen data.

The precision-recall curve, displayed in Figure 1 (a), evaluates the trade-off between precision and recall for the model across different thresholds. The area under the precision-recall curve (AUC-PR) is 0.87, indicating a good balance between precision and recall. The ROC curve in Figure 1 (b) illustrates the trade-off between the true positive rate and the false positive rate across different classification thresholds. The area under the ROC curve (AUC-ROC) is 0.86, indicating that the model has a strong ability to distinguish between driver and passenger indels.



(a) (b)

Figure 1: Performance curves for the XGBoost Classifier: (a) Precision-Recall and (b) ROC.

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